

Felix Saaro

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Machine Learning Engineer with a focus on evaluation and measurement in AI systems. Engineering is what excites me most, but working in research gives me the chance to ask why things work and why they might not. Across projects spanning LLM fine-tuning, vision pipelines, TTS, and RAG systems, I kept seeing the same problem: the field moves fast but measures poorly. That gap became my research direction. My semester project and thesis both focus on evaluation, combining empirical methods with engineering in an attempt to evaluate the AI systems we build.

I believe evaluation will become one of the defining challenges of applied AI, and I want to be working on it.

Skills & Languages

ML/AI

Data

Programming

Infrastructure

Languages

PyTorch, PEFT, vLLM, NVIDIA Triton, Bayesian Inference, LLM Evaluation

Prodigy, PostgreSQL, SLURM, Sensor Simulation

Python, Node.js

Docker, Kubernetes, CI & CD in Azure DevOps and GitHub

German (Fluent, Native), English (Fluent, C1)

Experience

Research Assistant — ZHAW, Centre for Artificial Intelligence

2023 – Present

Winterthur, Switzerland

Part Time (alongside MSc studies)

Built and operated infrastructure for running LLMs on an on-premise multi-node GPU cluster using NVIDIA Triton and vLLM, enabling large-scale model experimentation across research teams.

Led development of an LLM fine-tuning pipeline structuring 30k medical records into 574 hierarchical labels using only 2k annotated samples.

Developed evaluation methodology for AI systems as the focus of both my semester project and ongoing thesis, combining empirical and Bayesian methods to measure model behavior in ways that go beyond standard benchmarks.

Developed a multimodal perception pipeline integrating event cameras and LiDAR for real-time aerial object detection, including a simulation-based synthetic data generation pipeline for training and evaluation.

Published research at **EMNLP** and **IJPH**.

DevOps Engineer — Bühler Group AG

2019 – 2023

Uzwil, Switzerland

Part Time (alongside BSc studies)

Prototyped an early RAG system (*FastAPI*, *PostgreSQL*) to improve discoverability of engineering knowledge across 100 repositories.

Front-End Engineer — CanvasHeroes B.V.

2017 – 2019

Amsterdam, The Netherlands

Delivered multiple production web and mobile applications. Scaled a production platform from 100 to 10k concurrent users by introducing caching strategies and distributed database replication.

Apprentice / Software Developer — Online Consulting AG

2013 – 2017

Wil, Switzerland

Contributed to a web-based platform for industrial after-sales services within an international SCRUM team.

Education

MSc Data Science – ZHAW in Winterthur, Switzerland

2023 – Present

Thesis: On the Evaluation of RAG Systems: A Bayesian View Developing a Bayesian framework for evaluating RAG systems, focusing on measurement reliability and reproducibility across model configurations.

Semester Project: Do NOT Classify and Count: Hybrid Attribute Control Success Evaluation Developed a new evaluation methodology for attribute control in text generation, moving beyond simple classification-based metrics. *Final Grade: 6/6*

BSc Computer Science – ZHAW in Winterthur, Switzerland

2020 – 2023

Thesis: The Influence of Audio Length on the Performance of Swiss-German Speech Translation Models. *Final Grade: 5.5/6*

Exchange Semester: University of Agder, Grimstad, Norway.

Fall 2022

Publications

Do NOT Classify and Count: Hybrid Attribute Control Success Evaluation

EMNLP 2026

First Author – Peer-Reviewed, Published

AI Powered Patient Records Analysis for Injury Surveillance in Children and Adolescents

IJPH 2025

Second Author – Submitted

Projects

Skäirim - Swiss-German Voice Mod for Skyrim

Ongoing

Localizes game dialogue by cloning character voices using a fine-tuned Qwen3-TTS model conditioned on character race and dialect. Trained on 8,000 hours (3.5TB) of speech across seven Swiss-German dialects using multi-GPU pipelines. Includes dialect verification tooling to evaluate whether generated speech matches target dialect characteristics.

Bayesian Evaluation Framework for Agentic Pipelines

Ongoing

A framework for evaluating multi-step AI pipelines by modelling them as Bayesian networks, where each component's success rate depends on upstream systems. Unlike end-to-end evaluation, this approach surfaces where errors propagate and quantifies the impact of improving individual components — helping teams make informed decisions about where to invest.

Conditional Generation Evaluator

2025

A lightweight evaluation tool that aligns automatic classifier outputs with human judgment using only 100 annotations. Takes binary classifier predictions as input and produces a posterior distribution over the full dataset that reflects the human annotation distribution.

Volunteer Experience

President – Handballclub Uzwil

2024 – Present

Leading a club of 160 members across 17 teams, including organizing two regional youth tournaments with 400 participants each.